

Missing value Treatment

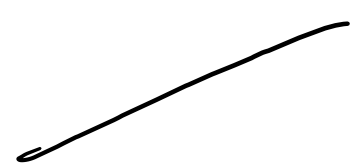
Missingno. py



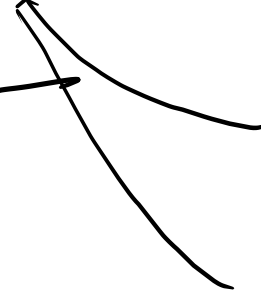
bar
matrix

heat map
dendrogram

Confirmatory tests



MCAR



Likelihood
MCMC
test

Dixon's

Hawkins's
test

global

$H_0: \text{MCAR}$
 $H_1: \text{NOT MCAR}$

α , two-tailed test

FDA
 Univariate
 Histogram
 Q-Q plots

p

Test Statistic

$\alpha = 0.05$
 S.I. $\chi^2_0 = y$
 $\chi^2_\alpha = x$

fail to reject

Reject
 $\alpha = 0.05$

$$p = 0.26$$
$$\alpha = 0.05$$

$$p > \alpha$$

→ Fail to Reject
Missing Data MCAR | H_0

$$p = 0.01$$
$$\alpha = 0.05$$

$$p < \alpha$$

→ Reject H_0
Not MCAR

MCAR

Treatment

% missing values

df. is small

* is na

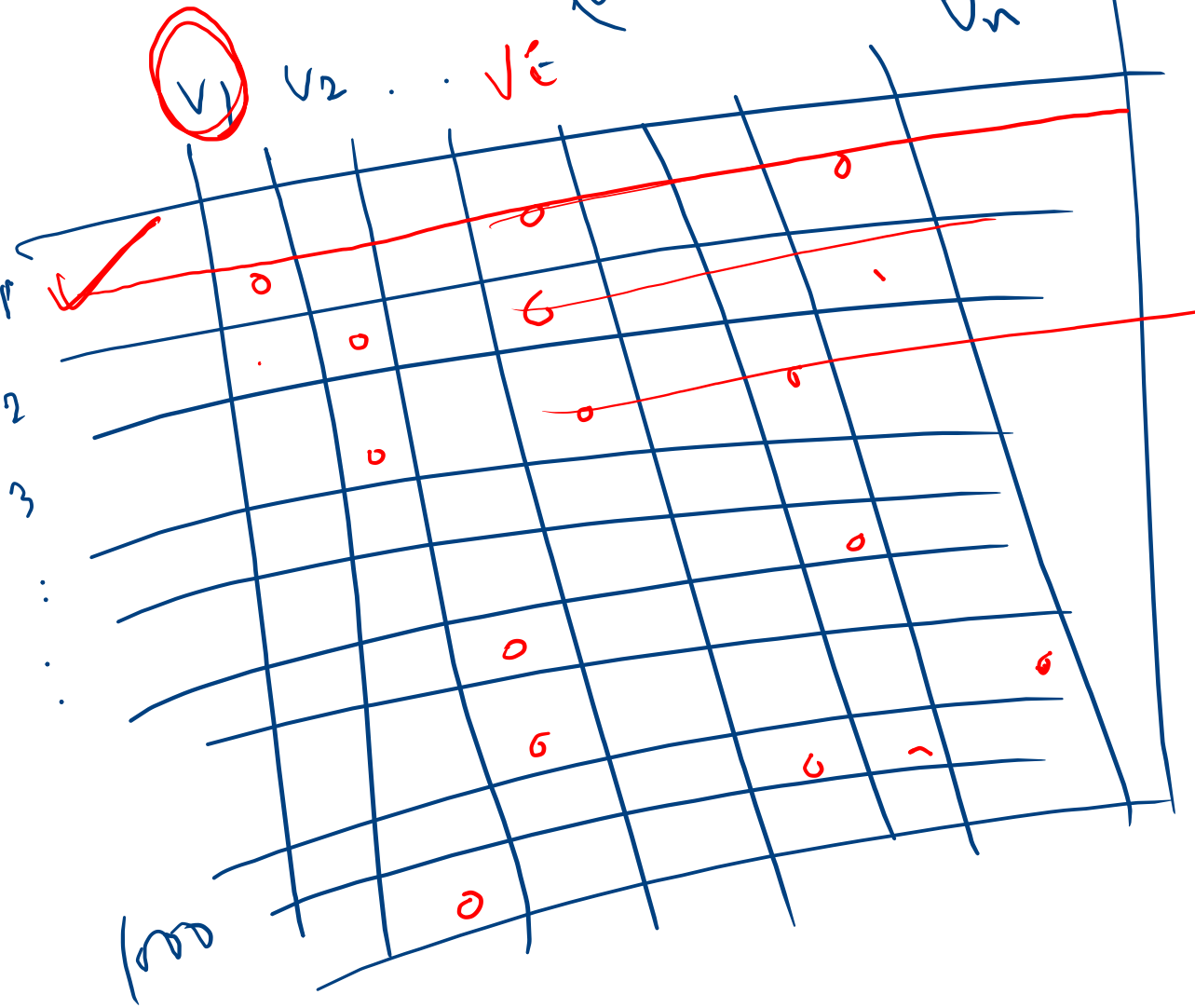
info

5-10%

Complete Case Analysis

Complete case
(list wise deletion)

Available case
(pairwise deletion)



Imputation

Mean

Median

function

Numeric
data

Regression

$$V_i = C_0 + C_1 V_2 + C_3 V_3 + E$$